1 EXAM PRACTICE

1 How many molecules of oxygen are there in a 1.00 g sample of O_2?

(Avogadro constant, $L = 6.02 \times 10^{23}$ mol^{-1})

A 1.88×10^{22}

B 3.76×10^{22}

C 9.63×10^{24}

D 1.93×10^{25} [1]

(Total for Question 1 = 1 mark)

2 What is the concentration, in mol dm^{-3}, of a solution of sodium chloride containing 4.27 g of NaCl in 300 cm^3 of solution?

A 0.0219

B 0.243

C 4.11

D 45.7 [1]

(Total for Question 2 = 1 mark)

3 The overall equation for one method of manufacturing hydrogen from methane is

$$CH_4 + 2H_2O \rightarrow 4H_2 + CO_2$$

In one batch, 5.12 tonnes of hydrogen were obtained from 13.6 tonnes of methane.

What is the percentage yield of hydrogen for this batch?

A 37.6

B 42.5

C 54.4

D 75.3 [1]

(Total for Question 3 = 1 mark)

4 The equation for a reaction that can be used to manufacture lithium carbonate is

$$2Li_2O_2 + 2CO_2 \rightarrow 2Li_2CO_3 + O_2$$

What is the atom economy for this reaction?

A 58.9

B 62.1

C 69.8

D 82.2 [1]

(Total for Question 4 = 1 mark)

5 The equation for a displacement reaction is

$$Zn + FeSO_4 \rightarrow Fe + ZnSO_4$$

Which species is displaced in this reaction?

A zinc

B iron

C sulfur

D oxygen [1]

(Total for Question 5 = 1 mark)

6 The equation for a precipitation reaction is

$$AgNO_3(aq) + NaCl(aq) \rightarrow AgCl(s) + NaNO_3(aq)$$

A solution containing 0.04 mol silver nitrate is added to a solution containing 0.06 mol sodium chloride.

What mass of precipitate forms?

A 5.74 g

B 7.17 g

C 8.60 g

D 14.3 g [1]

(Total for Question 6 = 1 mark)

7 Magnesium chloride may be prepared by reacting magnesium with dilute hydrochloric acid.

A sample of magnesium of mass 1.215 g was added to 60.0 cm^3 dilute hydrochloric acid of concentration 2.00 mol dm^{-3}.

The equation for the reaction is

$$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

(a) State two observations that are made when magnesium reacts with dilute hydrochloric acid. [2]

(b) (i) Calculate the amount, in moles, of magnesium used. [1]

(ii) Calculate the amount, in moles, of hydrochloric acid used. [1]

(iii) Use your answers to (b)(i) and (b)(ii) to show that the hydrochloric acid is in excess. [2]

(c) Calculate the volume, measured at r.t.p. of hydrogen produced in this reaction.

(The molar volume of hydrogen at r.t.p. is 24 dm^3 mol^{-1}) [2]

(Total for Question 7 = 8 marks)

8 Ammonium nitrate (NH_4NO_3) can be prepared by the reaction of ammonia with nitric acid.

The equation for the reaction is

$$NH_3(aq) + HNO_3(aq) \rightarrow NH_4NO_3(aq)$$

(a) (i) Calculate the minimum volume of nitric acid, of concentration 0.500 mol dm^{-3}, that is required to react completely with 25.0 cm^3 of aqueous ammonia of concentration 2.00 mol dm^{-3}. [2]

(ii) How could a solid sample of ammonium nitrate be obtained from the reaction mixture? [1]

(b) When ammonium nitrate is heated, it decomposes according to the following equation:

$$NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(l)$$

A 4.00 g sample of ammonium nitrate was carefully heated to produce only N_2O and water.

(i) Calculate the amount, in moles, of ammonium nitrate used. [2]

(ii) Calculate the volume of N_2O that was formed. [1]

(Assume that the molar volume of N_2O is $24\ dm^3\ mol^{-1}$ under the experimental conditions.)

(iii) N_2O can be decomposed into its elements by further heating. Write an equation for this reaction. Include state symbols. [2]

(Total for Question 8 = 8 marks)

9 Phosphorus forms three chlorides of molecular formulae PCl_3, PCl_5 and P_2Cl_4.

(a) State what is meant by molecular formula. [1]

(b) PCl_5 can be prepared by reacting white phosphorus (P_4) with chlorine gas.

Write an equation for this reaction. State symbols are not required. [1]

(c) Solid PCl_5 reacts vigorously with water to form a solution containing a mixture of two acids, H_3PO_4 and HCl.

Write an equation for this reaction. Include state symbols. [2]

(d) A compound was found to have the following percentage composition by mass:

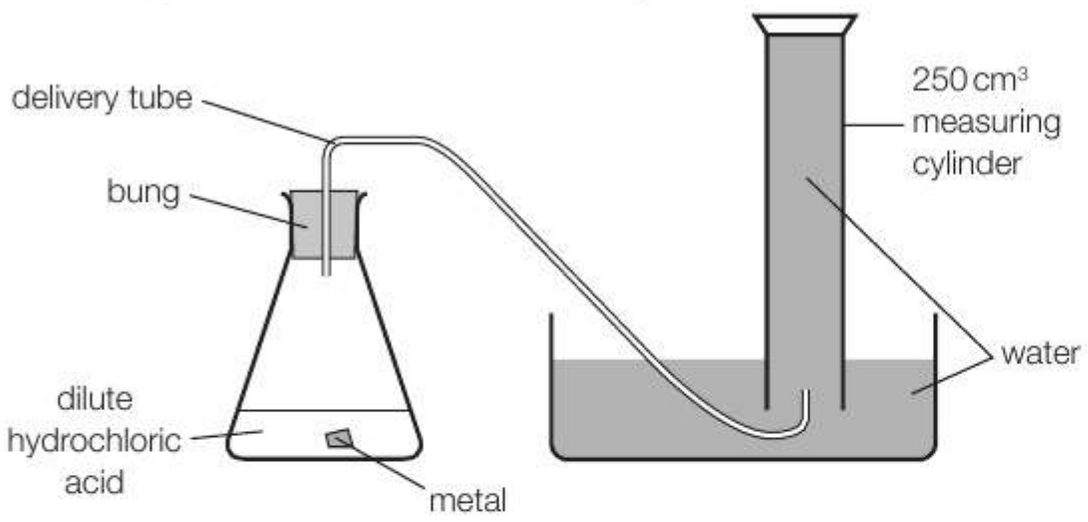
P = 30.39%; Cl = 69.61%

(i) Use this information to identify the chloride. [3]

(ii) Give the systematic name for this chloride. [1]

(Total for Question 9 = 8 marks)

10 The diagram shows the apparatus used to collect and measure the volume of hydrogen given off when a sample of a Group 2 metal reacts with dilute hydrochloric acid.



The equation for the reaction is

$$M(s) + 2HCl(aq) \rightarrow MCl_2(aq) + H_2(g)$$

where M is the symbol for the Group 2 metal.

The apparatus was set up as shown in the diagram.

A sample of the metal was weighed and then placed into the conical flask.

The bung was removed and an excess of acid was then added to the metal. The bung was replaced.

When the reaction was complete, and the gas collected had cooled to room temperature, the volume of gas collected in the measuring cylinder was measured.

The results are shown in the table.

Mass of metal	0.24 g
Volume of hydrogen collected	230 cm^3

(a) (i) Use the results to calculate the molar mass of the metal.

Assume the molar volume of hydrogen is $24.0\ dm^3\ mol^{-1}$ under the experimental conditions. [3]

(ii) Give the most likely identity of the metal. Justify your answer. [2]

(b) (i) Identity the major procedural error in the experiment. [1]

(ii) State a modification that would reduce this procedural error. [1]

(Total for Question 10 = 7 marks)

11 Azides are compounds of metals with nitrogen. Some azides are used as detonators in explosives. However, sodium azide (NaN_3) is used in air bags in cars.

(a) Sodium azide decomposes into its elements when heated.

$$2NaN_3(s) \rightarrow 2Na(l) + 3N_2(g)$$

What is the volume of gas produced, measured at r.t.p., when one mole of sodium azide is decomposed?

A 24 dm^3

B 36 dm^3

C 48 dm^3

D 72 dm^3

(Assume one mole of gas occupies 24 dm^3 at r.t.p.) [1]

(b) (i) A student completely decomposed 3.25 g of sodium azide. Calculate the mass of sodium she obtained. [2]

(ii) She then carefully reacted the sodium obtained with water to form 25.0 cm^3 of aqueous sodium hydroxide.

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

Calculate the concentration, in mol dm^{-3}, of the aqueous sodium hydroxide. [2]

(Total for Question 11 = 5 marks)